

Ankita Ghosh

Website $\diamond$ LinkedIn $\diamond$ GitHub $\diamond$ Google Scholar

Zurich, Switzerland



EDUCATION

ETH Zurich

MSc in Computer Science (Major in Visual and Interactive Computing)

Relevant Coursework: 3D Vision, Digital Humans, Machine Perception, Computer Graphics

September 2022 – June 2026

CGPA: 5.64/6.0

Manipal Institute of Technology

B.Tech in Computer Science and Engineering (Minor in Graphics and Visualization)

Relevant Coursework: Deep Learning, Digital Image Processing, Augmented and Virtual Reality

August 2018 – June 2022

CGPA: 9.27/10

EXPERIENCE

Teaching Assistant, ETH Zurich

Lecturer: Dr. Carlos Cotrini

June 2026 - Present

Zurich, Switzerland

- Assisting in course content formulation of lectures, exercises, and projects to ensure a comprehensive learning experience for the Master of Advanced Studies in AI and Digital Technology.

Visiting Graduate Scholar, ARCADE Lab, Johns Hopkins University

Supervisor: Dr. Lalithkumar Seenivasan and Prof. Dr. Mathias Unberath

July 2025 - March 2026

Baltimore, USA

- Developed a **neural rendering** pipeline for **dynamic digital twins**, integrating **6D pose estimation** to bridge the gap between physical operating rooms and simulation. *Under Review Nature Communications Journal*.
- Formulated a **scalable** grasp affordance **data generation framework** within a physics-based **simulation**, synthesising **dexterous grasps** from RGB-D images for surgical tool geometries. *Under Review CoRL 2026*.

Machine Vision Engineer Intern, Stryker

Sensing and Machine Vision Team

October 2024 – March 2025

Freiburg, Germany

- Developed a surgeon **tracking** and **keypoint detection** pipeline on operating room footage for surgical planning.
- Executed **cross-modal calibration** of RGB camera and optical trackers for **depth estimation** of the detected keypoints to design a deployable workflow for surgical workload estimation ultimately.

Student Researcher, VLG, ETH Zurich

Supervisor: Dr. Korrawe Karunratanakul and Prof. Dr. Siyu Tang

March 2023 - September 2024

Zurich, Switzerland

- Extended single-person generative models, integrating **contact-based annotations** and a **correlation model** to ensure physical plausibility in generated sequences.
- Architected a **transformer-based diffusion** pipeline for multi-agent motion synthesis, utilizing **vision language models** to align generated interactions with textual prompts.

Mitacs Globalink Research Intern, SIRRL, University of Waterloo

Supervisor: Prof. Dr. Moojan Ghafurian

June 2021 – September 2021

Ontario, Canada

- Developed an **emotion recognition system** that can process speech and respond with appropriate emotion in real-time by implementing the computational model of **affect control theory**.
- Designed facial expressions for **social robot Furhat** using **facial action coding system**, and developed a novel model that maps emotions to these facial gestures based on **semantic differential values**.

PUBLICATIONS

[1] H. Zhang, Y. Shen, R. D Soberanis-Mukul, **A. Ghosh**, H. Ding, L. Seenivasan, J. L Porras, Z. Mao, C. Li, W. Xiao, L. Yarmus, A. C. Argento, M. Ishii, M. Unberath, **TwinOR: Photorealistic Digital Twins of Dynamic Operating Rooms for Embodied AI Research**, *IPCAI 2026* [[paper](#)]

[2] **A. Ghosh**, S. Khose, Y. S. Kamath, Neetha I. R. Kuzhupilly, Harish Kumar J R, **Fovea Segmentation Using Semi-Supervised Learning**, *INDICON 2023* [[paper](#)]

[3] S. Khose, **A. Ghosh**, Y. S. Kamath, Neetha I. R. Kuzhupilly, Harish Kumar J R, **Explainable Classification of Macular Degeneration Using Deep Learning**, *INDICON 2023* [[paper](#)]

[4] R. Naidu, **A. Ghosh**, Y. Maurya, S. R. Nayak, S. S. Kundu, **IS-CAM: Integrated Score-CAM for axiomatic-based explanations**, *Responsible Computer Vision, CVPR-W 2021* [[paper](#)]

- [5] A. Ghosh*, S. Khose*, A. Tiwari*, **Semi-Supervised Classification and Segmentation on Aerial Images, Tackling Climate Change with Machine Learning, NeurIPS-W 2021** [[paper](#)]
- [6] V Manushree*, S. Saxena*, P. Chowdhury*, M. Varma*, H. Rathod*, A. Ghosh, S. Khose, **Extraction of Color Information from Images for Generation of Colored-Sketches, Machine Learning for Creativity and Design, NeurIPS-W 2021** [[paper](#)]

RELEVANT PROJECTS

- How Much Noise is Too Much Noise?** Spring 2024
Computational Semantics for NLP Course Project at ETH Zurich [[code](#) | [report](#)]
- Implemented an alignment pipeline using reinforcement learning from human feedback (**RLHF**) and **preference optimization** to evaluate model robustness against noisy annotations.
 - Benchmarked PPO, DPO and N-Sampling performance by analyzing **reward model evaluations** and **KL-divergence** trade-offs from the reference language model.
- Latent Space Exploration for Generative AI** Spring 2024
Interactive Machine Learning Course Project at ETH Zurich [[report](#) | [poster](#)]
- Designed an **interactive framework** for exploration and manipulation of the learned representations of a **generative model** trained on textile data.
 - Produced 2D projection of **disentangled latent** space along colors to enable **user-controlled** textile generation.
 - Automated the deployment lifecycle through a **CI/CD** pipeline and **Docker** containerization.
- Leveraging Motion Imitation in Reinforcement Learning for Biped Character** Spring 2023
Digital Humans Course Project at ETH Zurich [[code](#) | [report](#) | [demo](#)]
- Implemented an **actor-critic algorithm** that performs **task objectives** like direction control, alongside imitating motions in a physically-based environment.
 - Synthesized longer motion sequences by using methods like **multi-clip concatenation** and **composite policy**.
 - Proposed a **residual policy network** that can leverage pre-trained agents and retarget to new characters.
- Semantic-MD: Infusing Monocular Depth with Semantic Signals** Spring 2023
3D Vision Course Project at ETH Zurich [[code](#) | [report](#) | [poster](#)]
- Explored different ways of integrating semantic signals to the image input through concatenation and convolutions in the form of **one-hot encoded semantic maps and contours**.
 - Performed **multi-task learning** to jointly estimate depth and semantic maps.
 - Conducted extensive **ablation studies** with different segmentation architectures and loss functions where depth estimation achieved a decrease of **12.86%** in relative mean error with the aid of semantic information.
- Scene Render: Man on Mars** Autumn 2022
Computer Graphics Course Project at ETH Zurich [[report](#)]
- Implemented a **physically-based renderer** with **light source functionalities** like environment map emitter and probabilistic progressive photon mapping, and additional post-processing NL-means **denoising**.
 - Modeled surface appearances** by overlaying image textures, normal mapping, and reproducing **Disney BRDF**.

TECHNICAL SKILLS

Languages: Python, C, C++, Java, SQL, Kotlin, JavaScript

Expertise: 3D Vision, Healthcare AI, Generative AI, Multimodal Learning, Explainable AI, Interactive Systems

Tools and Frameworks: PyTorch, OpenCV, Isaac Lab, Blender, Slicer, Docker, Git, mySQL

EXTRACURRICULAR

- Co-Founder and Technical Head, [Research Society Manipal](#):** administered a student body of 100+ members promoting inter-disciplinary research, mentored students in the field of AI, and hosted academic events like webinars.
- Volunteer, [Teach Code for Good, Manipal](#):** tutored 20 underprivileged students in secondary school on Computer Science topics and programming languages like Python and C.
- Writer, [Manipal The Talk Network](#):** published 10+ articles constituting informative features on technology and creative literary pieces at Manipal's largest independent media organization in India.